

MATH20250 HW4 Observations

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Problem-Specific Observations

I again have solutions to problems for which I identified areas or concepts with explanations worth repeating. As always, consult the debrief file on Canvas and the compiled example solutions from your fellow students.

LADW 3.3.10. Let A be a square matrix. Show that block triangular matrices

$$\begin{pmatrix} I & * \\ \mathbf{0} & A \end{pmatrix}, \begin{pmatrix} A & * \\ \mathbf{0} & I \end{pmatrix}, \begin{pmatrix} I & \mathbf{0} \\ * & A \end{pmatrix}, \begin{pmatrix} A & \mathbf{0} \\ * & I \end{pmatrix}$$

all have determinant equal to $\det(A)$. Here $*$ can be anything.

I will address the first of these matrices in the greatest detail; hopefully, it will be clear how the approach can be adapted to each of the other cases.

Suppose you have the block triangular matrix $\begin{pmatrix} I & * \\ \mathbf{0} & A \end{pmatrix}$, where I is the $m \times m$ identity matrix and A is an $n \times n$ matrix, with $*$ and $\mathbf{0}$ then dimensioned appropriately for the overall matrix to be $(m+n) \times (m+n)$. Consider A by itself for a moment. We know that A is row-equivalent to a matrix in *echelon* form – we can obtain such a matrix by performing elementary row operations on A . In this case, we note that we can find a particular upper triangular matrix that is row-equivalent to A .

Note: Strictly speaking, we can perform any of the elementary row operations – scaling, swapping, and adding a multiple of one row to another – on A to obtain an upper triangular matrix. The important thing is to keep track of what each transformation does to the determinant in order to scale the determinant of the matrix you obtain later properly. For example, we have

$$\begin{pmatrix} 3 & 2 & 1 \\ 3 & 6 & 3 \\ 0 & 1 & -1 \end{pmatrix} \xrightarrow{r_2 \rightarrow \frac{1}{3}r_2} \begin{pmatrix} 3 & 2 & 1 \\ 1 & 2 & 1 \\ 0 & 1 & -1 \end{pmatrix} \xrightarrow{r_1 \leftrightarrow r_2} \begin{pmatrix} 1 & 2 & 1 \\ 3 & 2 & 1 \\ 0 & 1 & -1 \end{pmatrix} \xrightarrow{r_2 \rightarrow r_2 - 3r_1} \begin{pmatrix} 1 & 2 & 1 \\ 0 & -4 & -2 \\ 0 & 1 & -1 \end{pmatrix} \\ \xrightarrow{r_2 \leftrightarrow r_3} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \\ 0 & -4 & -2 \end{pmatrix} \xrightarrow{r_3 \rightarrow r_3 + 4r_2} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & -6 \end{pmatrix}.$$

So, the initial and final matrices are row-equivalent to each other, but the intervening operations have yielded two matrices with differing determinants. In particular, the determinant changed with the $\frac{1}{3}$ row-scaling, the swap of r_1 and r_2 , and the swap of r_2 and r_3 . We then have ¹

$$\begin{vmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & -6 \end{vmatrix} = \left(\frac{1}{3}\right) \cdot (-1) \cdot (-1) \begin{vmatrix} 3 & 2 & 1 \\ 3 & 6 & 3 \\ 0 & 1 & -1 \end{vmatrix}.$$

Moving back to the larger block matrix, note that the final n rows contain precisely zeroes and the elements of A . So, you can perform row operations that affect **only those rows** to get the A block in triangular form. At that point, you need only to have kept track of any scaling from the row operations, as the determinant of the larger matrix will be scaled by precisely those factors. Note, then, that once A is in (upper) triangular form, the product of those n diagonal elements (along with any scaling factors) is precisely $\det(A)$.

As far as the larger matrix's determinant, note that the first m rows are already in (upper) triangular form by virtue of the positioning and structure of the identity matrix. So, the determinant of the full – now triangular – matrix is the product of all the diagonal elements and scaling factors from the A block row operations. Since the first m diagonal elements are all equal to 1, this necessary means that

$$\begin{vmatrix} I & * \\ \mathbf{0} & A \end{vmatrix} = \det(A).^2$$

We can follow the same procedure to prove the same for the other three matrices with the following adjustments:

1. $\begin{pmatrix} A & * \\ \mathbf{0} & I \end{pmatrix}$: The only difference is that the echelon form of A will take up the first n rows (and columns) instead of the latter n . Once more, $*$ does not matter for our purposes.
2. $\begin{pmatrix} I & \mathbf{0} \\ * & A \end{pmatrix}$: We just make sure the echelon form of A is lower triangular.
3. $\begin{pmatrix} A & \mathbf{0} \\ * & I \end{pmatrix}$: Again, this is just the lower triangular version of the second case.

Therefore, we have that

$$\begin{vmatrix} I & * \\ \mathbf{0} & A \end{vmatrix} = \begin{vmatrix} A & * \\ \mathbf{0} & I \end{vmatrix} = \begin{vmatrix} I & \mathbf{0} \\ * & A \end{vmatrix} = \begin{vmatrix} A & \mathbf{0} \\ * & I \end{vmatrix} = \det(A). \quad \square$$

¹Recall that the vertical bars replacing the matrix brackets/parenthesis signal the determinant.

²Note how $*$ was indeed immaterial here.

LADW 3.4.1. Suppose the permutation σ takes $(1, 2, 3, 4, 5)$ to $(5, 4, 1, 2, 3)$.

(a) Find the sign of σ .

It is helpful to write the permutation in cycle notation; that is, we essentially group the permuted “letters”³ with their “partners.” A *cycle* is a subset of the overall set of letters such that, when the permutation is applied to the whole set, the letters in the cycle are always permuted *specifically* with each other. In this case, we have the following:

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 1 & 2 & 3 \end{pmatrix} = (1 \ 5 \ 3) (2 \ 4).$$

We arrive at these cycles by “following a thread” and seeing

$$\sigma(1) = 5, \sigma(5) = 3, \sigma(3) = 1 \Rightarrow (1 \ 5 \ 3)$$

$$\sigma(2) = 4, \sigma(4) = 2 \Rightarrow (2 \ 4).$$

However, in order to find the sign of the permutation, we need to decompose the cycles further into *transpositions*: cycles of length 2. Here, we can split up the three cycle and obtain the following product of transpositions:

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 1 & 2 & 3 \end{pmatrix} = (1 \ 5 \ 3) (2 \ 4) = (1 \ 5) (5 \ 3) (2 \ 4).$$

Since there are three transpositions in this permutation, we have that $\boxed{\text{sign}(\sigma) = -1}$.

(b) What does $\sigma^2 := \sigma \circ \sigma$ do to $(1, 2, 3, 4, 5)$?

Returning to the less-decomposed cycle notation, we find a very helpful visual aid. Recall that σ is given by

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 1 & 2 & 3 \end{pmatrix} = (1 \ 5 \ 3) (2 \ 4).$$

Start with 1. In order to see what $\sigma^2(1)$ is, we start at 1 in its 3-cycle and move two spots to the *right*:

$$(1 \ 5 \ 3) \rightarrow \left(\underline{1} \quad \underset{\sigma(1)}{5} \quad 3 \right) \rightarrow \left(\underline{1} \quad \underset{\sigma(1)}{5} \quad \boxed{\underset{\sigma^2(1)}{3}} \right) \Rightarrow \sigma^2(1) = 3.$$

Moving now to $\sigma^2(2)$, we start at 2 in its 2-cycle and count off two spots to the right, wrapping around to the beginning if necessary:

$$(\underline{2} \ 4) \rightarrow \left(\underline{2} \quad \underset{\sigma(2)}{4} \right) \rightarrow \left(\boxed{\underset{\sigma^2(2)}{2}} \quad \underset{\sigma(2)}{4} \right) \Rightarrow \sigma^2(2) = 2.$$

Repeating for the remaining letters, we have

$$\boxed{\sigma^2(1, 2, 3, 4, 5) = (3, 2, 5, 4, 1)}.$$

³Often, the group S_n of permutations of the first n positive integers is called the “group of permutations on n letters.”

(c) **What does the inverse permutation σ^{-1} do to $(1, 2, 3, 4, 5)$?**

Using the same cycle notation as in the previous part, we now will start at a letter and move one spot to the *left*, wrapping around to the “end” of the cycle if necessary. For example, we have the following for 1:

$$(\underline{1} \ 5 \ 3) \rightarrow \left(\underline{1} \ 5 \ \boxed{\begin{smallmatrix} 3 \\ \sigma^{-1}(1) \end{smallmatrix}} \right) \Rightarrow \sigma^{-1}(1) = 3.$$

Repeating for the remaining letters, we then have

$$\boxed{\sigma^{-1}(1, 2, 3, 4, 5) = (3, 4, 5, 2, 1)}.$$

(d) **What is the sign of σ^{-1} ?**

Note that, in applying the inverse permutation, we essentially just apply the transpositions “in reverse.” We then see

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 2 & 1 \end{pmatrix} = (1 \ 3 \ 5)(2 \ 4) = (1 \ 3)(3 \ 5)(2 \ 4).$$

Since we perform three transpositions, we therefore have that $\boxed{\text{sign}(\sigma^{-1}) = \text{sign}(\sigma) = -1}$.

Side note: What about $\text{sign}(\sigma^2)$? Here is the cycle notation for σ^2 :

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 5 & 4 & 1 \end{pmatrix} = (1 \ 3 \ 5)(2)(4) = (1 \ 3 \ 5)^4 = (1 \ 3)(3 \ 5).$$

So, since there are only really two transpositions that occur, we see that $\text{sign}(\sigma^2) = 1$. In practice, there are $2 \times 3 = 6$ transpositions, but when the dust settles, there are only effectively two transpositions that have taken place.

⁴1-cycles, which necessarily denote static elements, need not be included explicitly in cycle notation.

LADW 3.7.2. Let A be an $n \times n$ matrix. How are $\det(3A)$, $\det(-A)$, and $\det(A^2)$ related to $\det(A)$?

Recall that *multiplying a single row or column of a matrix by a scalar scales the determinant by that scalar*. So, what if you multiply the *entire matrix* by a scalar?

Since A is an $n \times n$ matrix, $3A$ represents the matrix given by multiplying *each of the n rows* of A by 3. As such, we have $\det(3A) = 3^n \det(A)$.

Similarly, $-A$ represents the matrix given by multiplying each of the n rows of A by -1 . So, $\det(-A) = (-1)^n \det(A)$.

IMPORTANT: In general, $(-1)^n \neq -1^n$. For all $n \in \mathbb{N}$, $-1^n = -1$, whereas $(-1)^n = 1$ for even n and -1 for odd n . *Please use parentheses where necessary.*

Finally, by Theorem 3.5 in LADW, $\det(A^2) = \det(A \cdot A) = \det(A) \cdot \det(A) \Rightarrow \det(A^2) = (\det(A))^2$.

Supp. III. Let $f(x) = \det(A)$ where

$$A = \begin{pmatrix} -6 & 9 & -2 & 1 & 3 \\ 4 & 0 & 7 & 1 & -5 \\ 4 & 0 & 0 & -8 & 8 \\ x & 9 & -3 & -8 & -9 \\ 4 & 0 & 0 & 0 & -2 \end{pmatrix}.$$

Find $f'(x)$.

Recall that $f'(x)$ is the first derivative of the function $f(x)$ (with respect to x). Consider what the determinant of A here represents. In particular, note that cofactor expansion along the first column amounts to a sum of four constant determinants and the variable x multiplied by a constant determinant.

Essentially, $f(x) = \det(A)$ is an expression that is linear in x . This means that $f'(x)$ is simply the coefficient of x in $f(x)$, as the derivative is itself a linear transformation and the derivative of constants is 0. So, we have

$$f'(x) = (-1)^{4+1} \det(A_{4,1}) = - \begin{vmatrix} 9 & -2 & 1 & 3 \\ 0 & 7 & 1 & -5 \\ 0 & 0 & -8 & 8 \\ 0 & 0 & 0 & -2 \end{vmatrix}.$$

As luck would have it, this matrix is upper triangular! This means its determinant is the product of its diagonal entries; we therefore have

$$f'(x) = -(9) \cdot 7 \cdot (-8) \cdot (-2) \Rightarrow f'(x) = -1008.$$

General Observations

IMPORTANT: Any problems with this boxed “x” next to them – **X** – should NOT be chosen for your HW submission. If you do this on one HW assignment, you will be warned but given credit. On subsequent assignments, you will not receive credit for problems that have been so marked.

- **Observations on Proofs**

- (a) **Directionality:** Suppose you have to prove “ A if and only if B .” ($A \Leftrightarrow B$) This is a **twofold** process:
 - i. **Prove “if A , then B .”** This may be done by supposing A is true and proving B directly, supposing A is true but B is false and proving by contradiction, or by contrapositive – “if A , then B ” is logically equivalent to “if not B , then not A .”
 - ii. **Prove “if B , then A .”** This may be done by supposing B is true and proving A directly, supposing B is true but A is false and proving by contradiction, or by contrapositive – “if B , then A ” is logically equivalent to “if not A , then not B .”
- (b) **Defining variables and other quantities:** Whenever you introduce a new quantity, be it a scalar, vector, etc., **state the space in which it lives.**

Ex. $\mathbf{v} \in V$ (where V is a vector space), $\alpha \in \mathbb{C}$.

The $\in$ symbol, meaning “is an element of,” is generated by `\in`.

- (c) **Existence:** If you are tasked with proving the existence of a quantity, **do not presuppose its existence in your notation.**
 - i. Example: LADW 1.6.7 – The goal is to prove that the matrix B is invertible given that the matrices A and AB are invertible. It may be tempting at some point in the proof to use the following fact from Theorem 6.3 (pg. 26):

$$(AB)^{-1} = B^{-1}A^{-1}.$$

However, there is a key condition attached to that fact:

If linear transformations A and B are **invertible**...

As such, this fact should not be employed unless you have already shown that A and B are invertible.

- (d) **Justifications:** There are two main things to remember when justifying the steps in a proof:
 - i. **Prove all facts on which you rely.** If you need to use a result that you did not already prove in class, **prove it explicitly.** If you need to use a result that comes from another homework problem, **prove it explicitly.** Do not take other homework problems for granted – see LADW 3.3.10-3.3.12.
 - ii. If you are unsure whether to justify something explicitly, **do it.** Better safe than sorry.

- Miscellaneous

- (a) **Check your calculations.** Especially beware of dropping negative signs and accidentally performing the wrong arithmetic operation on two numbers (e.g., adding instead of multiplying).
 - i. If you are finding a solution to a linear system of equations, **substitute your answer to check**. Sometimes, after converting to matrix form, it can be difficult to remember that there was an underlying group of equations. Use them to your advantage.
 - ii. If you are decomposing a matrix (e.g., writing it as a product of elementary matrices), **multiply the resulting matrices** to make sure you get the result you expect.
- (b) **Read the problems carefully.** Make sure you answer **every part of the problem**.
 - i. This also means you should not make any additional assumptions than what the problem allows.
 - ii. Additionally, for true/false questions, if a statement is true under a certain set of circumstances but *not under the set of circumstances outlined* in the statement, **the statement is false**. If no more specifics are given, err on the side of generality.